

Problemas Integrales Indefinidas.

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Problema 1 Calcular las siguientes integrales:

1. $\int (6x^2 + 8x + 3) dx$

2. $\int (\sqrt{x} + 1) (x - \sqrt{x} + 1) dx$

3. $\int \frac{3x^2 - 3}{x^3 - 3x} dx$

4. $\int \frac{x^2 + 5x + 7}{x + 3} dx$

5. $\int \frac{x}{\sqrt{x^2 + 1}} dx$

6. $\int \frac{\sqrt{x} + \ln x}{x} dx$

7. $\int (x - 1) \sqrt{x^2 - 2x} dx$

8. $\int a^{\sin x} \cos x dx$

9. $\int \frac{(e^{\arctan x} + x \ln(1 + x^2) + 1)}{1 + x^2} dx$

10. $\int \frac{6e^x}{\sqrt{1 - e^{2x}}} dx$

11. $\int (x + 1) e^{-x^2 - 2x - 5} dx$

12. $\int \frac{\ln x}{x^3} dx$

13. $\int x^2 e^{3x} dx$

14. $\int (x^2 - 2x + 5) e^{-x} dx$

15. $\int (x^2 + 5x + 6) \cos 2x dx$

16. $\int \arcsen x dx$

17. $\int \frac{dx}{x^3 - 2x^2 + x}$

18. $\int \frac{x^2 + 2x - 3}{(x - 1)^2 (x^2 + 1)} dx$

19. $\int \frac{x + 1}{x^2 + 3x + 3} dx$

20. $\int \frac{x^2}{\sqrt{4 - x^2}} dx$

21. $\int \sqrt{9 - 4x^2} dx$

22. $\int \frac{dx}{\sqrt{4x + x^2}}$

23. $\int \frac{xdx}{\sqrt{(1 - x^4)^3}}$